Homework Assignment 3

DPI 610

Assignment Details

This homework assignment is due *March 30, 2026* before midnight. Everyone must complete their own code and assignment, though you may discuss the homework with classmates.

To access the code and data for the assignment, go to Posit.cloud or Canvas. To turn in the assignment, upload two files to Canvas: (1) your code file (.qmd), and (2) a PDF version (click Render to create PDF).

```r
pkgs <- c("sf", "tidyverse", "tigris", "geomander", "spdep",
          "redist", # optional, for Q6 only
          "kableExtra", "urbnthemes")

check_pkgs <- function(pkgs) {
  for (pkg in pkgs) {
  if (!requireNamespace(pkg, quietly = TRUE)) {
    install.packages(pkg)
  }
  library(pkg, character.only = TRUE)
  }
}

check_pkgs(pkgs)
sf_use_s2(FALSE)  # Use planar geometry; avoids edge cases with s2 spherical geometry

## load data
load("hw3.RData")
```

Introduction

In this assignment, you will be analyzing spatial data in Massachusetts. We will begin by exploring some characteristics of local towns and then we will expand to study the entire state.

We have loaded three different local spatial data objects: `local_tracts` (census tract data with variables `inc` for *household median income* and `pop` for *population*), `local_votes` (precinct data with `dvoteshare` for 2020 Presidential *Democratic vote share*), and `local_boundary` (shapefile boundary for local towns). We have also loaded `mainroads`, which contains line geometries for two major roads in the area (Massachusetts Avenue and Alewife Brook Parkway).

For later in the problem set we also include `tracts` (the same census tract data for the full state) and `votes` (the same election data for the whole state, with variables `pre_20_dem_bid`, `pre_20_rep_tru`, `dvoteshare`, and `pop`).

The plot below illustrates census tract level median household income in our local area.

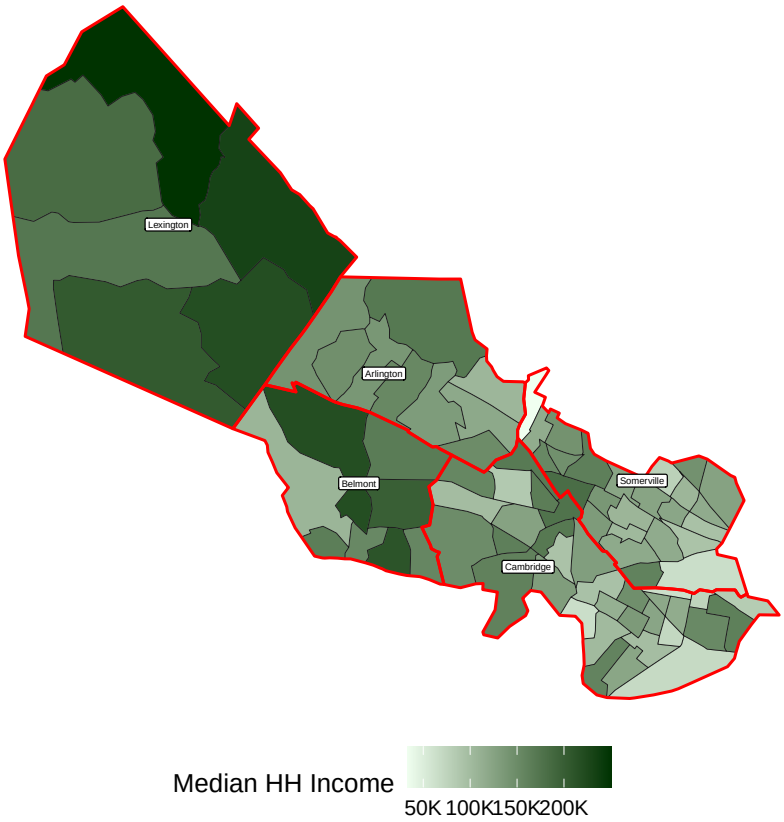

Figure 1: Local Median HH Income by Census Tract

Geography

Question 1

What town has the highest and lowest median household income census tract? (Hint: To figure out which tracts are in which town, you can use spatial functions such as `st_join()` in tandem with `st_buffer()`, or you may plot it and identify them visually.) Use the code for the plot above and modify it to highlight the wealthiest and poorest tracts on the map. Briefly comment on the spatial pattern of income in the area.

Answer 1

```
sf1 <- st_join(local_tracts, local_boundary,
               largest = TRUE) #' if tract touches two towns,
                               #' assign to town that contains more area
# highest HH income tract
sf1 %>% arrange(desc(inc)) %>%
  head(n = 1) %>% select(NAME, GEOID.x, inc) -> q1a

poly1 <- local_tracts %>% filter(GEOID == q1a[[2]])

# lowest HH income tract
sf1 %>% arrange((inc)) %>%
  head(n = 1) %>% select(NAME, GEOID.x, inc) -> q1b

poly2 <- local_tracts %>% filter(GEOID == q1b[[2]])

# map highest/lowest census tracts
ggplot() +
  geom_sf(data = local_tracts) +
  geom_sf(data = local_boundary, color = "black", fill = NA, linewidth = 0.6) +
  geom_sf(data = poly1, fill = "green", color = "black") +
  geom_sf(data = poly2, fill = "red", color = "black") +
  geom_sf_label(data = local_boundary, aes(label = NAME),
                size = 1.5,
                color = "black",
                fill = "white") +
  theme_void() +
  theme(legend.position = "bottom")
```

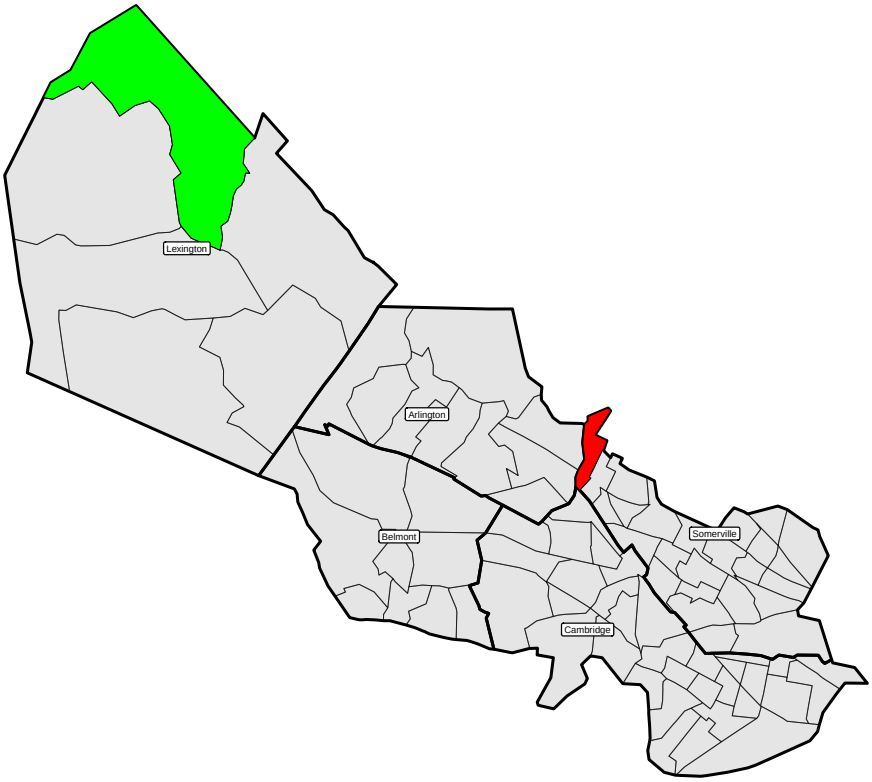

Figure 2: Highest (Green) and Lowest (Red) Median HH Income Census Tracts

- Lexington is the town with the highest median household income census tract (GEOID: 25017358700).
- Somerville is the town with the highest median household income census tract (GEOID: 25017350702).

Question 2

The "spatial lag" of a variable, is the *weighted average of that variable's values in neighboring geographic units*. Create a plot illustrating the relationship between each census tract's median household income and the spatial lag of median household income. What does this tell you about the level of spatial autocorrelation of this statistic? Furthermore, what is the Moran's I in this case?

Answer 2

```
#' for each census tract
  #' find neighboring tracts
    #' create spatial weights
    #' compute the spatial lag

# https://www.paulamoraga.com/book-spatial/spatial-neighborhood-matrices.html
# https://mgimond.github.io/simple_moransI_example/

# ---- Step 1: Define neighboring census tracts ----

#' construct a list of neighbors based on areas with contiguous boundaries
#' poly2nb() accepts a list of tracts and
#' returns a list of class nb with the neighbors of each area
nb <- poly2nb(local_tracts, queen = TRUE)
# head(nb)

#' plot the tracts map and overlap the neighborhood structure with the
#' plot.nb() function passing the neighbor list and the coordinates of the map
plot(st_geometry(local_tracts), border = "lightgray")
plot.nb(nb, st_geometry(local_tracts), add = TRUE)
```

Figure 3: Census Tracts Network

```
#' We can plot the neighbors of a given area
#' by adding a new column in map representing the neighbors of the area
cards <- card(nb) # number of neighbors per tract
max_id <- which.max(cards) # tract with the most neighbors

local_tracts$neighbors <- "other"
local_tracts$neighbors[max_id] <- "area"
local_tracts$neighbors[nb[[max_id]]] <- "neighbors"
```

```
# ---- Step 2: Assign weights to the neighbors ----

#' Each neighboring tract will be assigned equal weight when
#' computing the neighboring mean income values.
#'
```

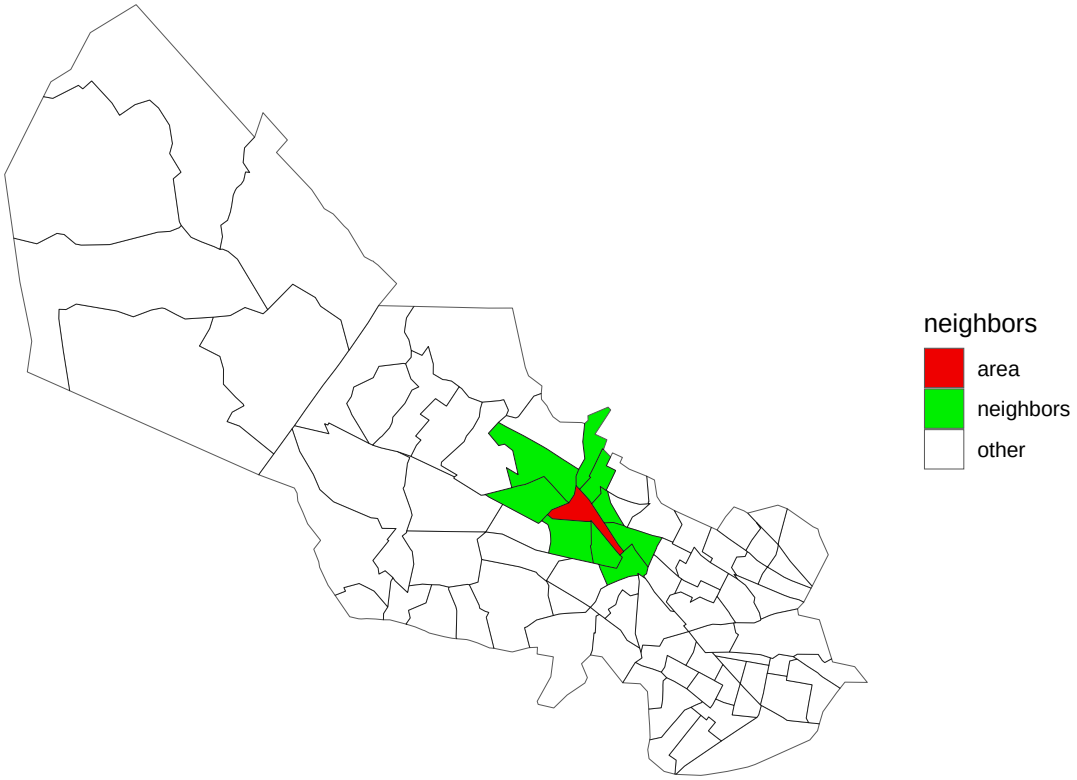

Figure 4: Census Tracts with the Most Neighboring Tracts (9)

```r
#' To compute a spatial weights matrix based on a binary neighbor list,
nbw <- nb2listw(nb, # list with neighbors
                style = "W", #' coding scheme chosen
                zero.policy = TRUE) # take into account polys with 0 neighbors

# see the weight of the first three tracts's neighbors
  # nbw$weights[1:3]
  #' These are the weights each neighboring income value
  #' will be multiplied by before being summed
  #' If a polygon has 5 neighbors, each neighbor will have a weight of 1/5 or 0.2
  #' This is equivalent to summing all five income values then dividing by 5

# ---- Step 3: Compute the weighted neighbor mean income values ----

#' Compute the average neighbor income value for each polygon
#' These are the spatially lagged values
#' Spatial Lag
Vx <- lag.listw(nbw, local_tracts$inc)
df <- data.frame(income = local_tracts$inc,
                 lag_income = Vx)

ggplot(df, aes(x = income, y = lag_income)) +
  geom_point(shape = 19) +
  geom_smooth(method = "lm", color = "red", se = TRUE) +
  labs(
    x = "Median Household Income",
    y = "Spatial Lag of Income"
  ) +
  theme_minimal()
```

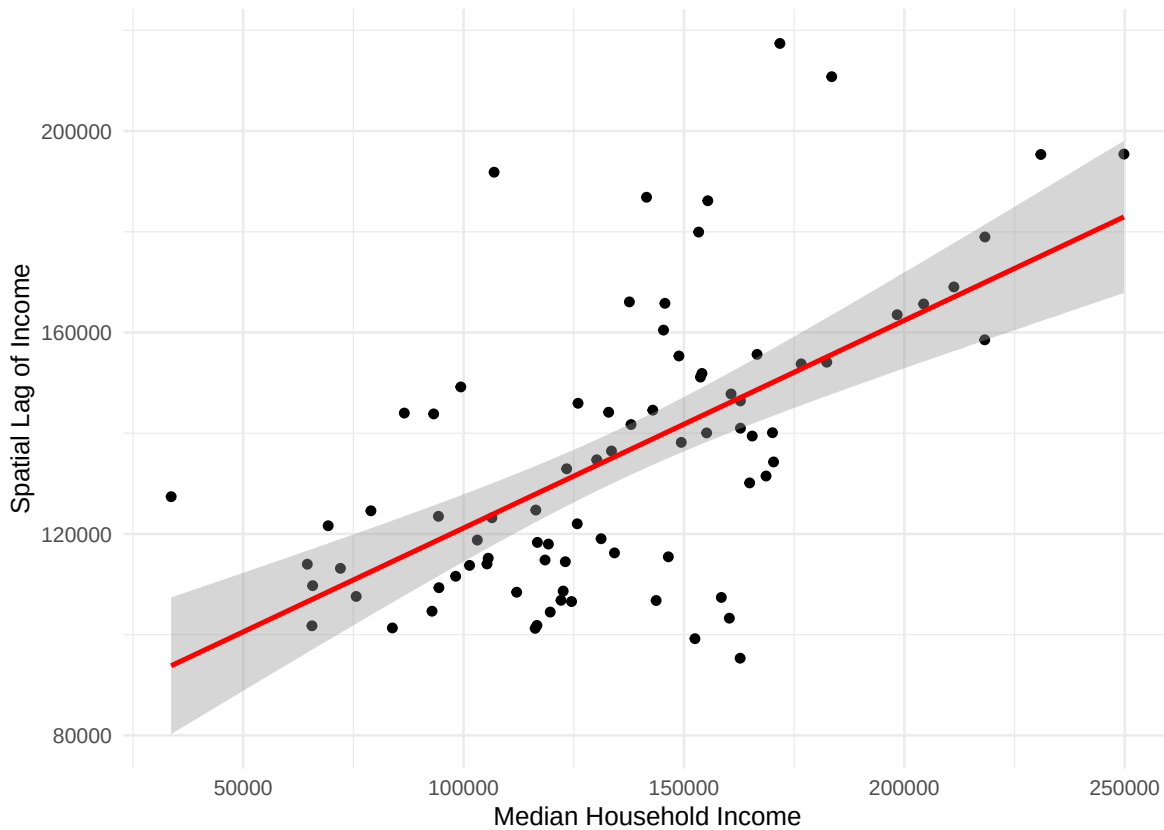

Figure 5: Association between the Mean HH Income of Tract and the Average MHI of Neighboring Tracts.

```
# Fit the linear model
model <- lm(lag_income ~ income, data = df)

# ---- Step 4: Computing the Moran's I statistic ---
moran <- moran.test(local_tracts$inc, nbw)
moran
```

```
    Moran I test under randomisation

data:  local_tracts$inc
weights: nbw

Moran I statistic standard deviate = 6.1869, p-value = 3.069e-10
```

```
alternative hypothesis: greater
sample estimates:
Moran I statistic       Expectation          Variance
      0.412331267      -0.012500000       0.004715114
```

- The above scatter plot illustrates a positive spatial autocorrelation. This is affirmed by the moran's I coefficient of 0.4123 which is positive, suggesting that as the average household income of any given census tract increases, so does the average household income of it's neighboring tracts.

Question 3

Repeat Q1 and Q2 but using the `local_votes` data and Democratic vote share (the `dvoteshare` variable). Is there more clustering for Democratic vote share or for median household income? Interpret the meaning of your finding.

Answer 3

```
# plot(st_geometry(local_votes))
sf_votes <- st_join(local_votes, local_boundary,
               largest = TRUE) #' if voter precinct touches two towns,
                               #' assign to town that contains more area

# plot(sf_votes)

# highest dem vote share precinct
sf_votes %>% arrange(desc(dvoteshare)) %>%
  head(n = 1) %>% select(NAME, GEOID20, dvoteshare) -> q3a

poly1 <- local_votes %>% filter(GEOID20 == q3a[[2]])

# lowest dem vote share precinct
sf_votes %>% arrange((dvoteshare)) %>%
  head(n = 1) %>% select(NAME, GEOID20, dvoteshare) -> q3b

poly2 <- local_votes %>% filter(GEOID20 == q3b[[2]])

# map highest/lowest precincts
ggplot() +
  geom_sf(data = local_votes) +
  geom_sf(data = local_boundary, color = "black", fill = NA, linewidth = 0.6) +
  geom_sf(data = poly1, fill = "blue1", color = "black") +
  geom_sf(data = poly2, fill = "purple2", color = "black") +
  geom_sf_label(data = local_boundary, aes(label = NAME),
                size = 1.5,
                color = "black",
                fill = "white") +
  theme_void() +
  theme(legend.position = "bottom")
```

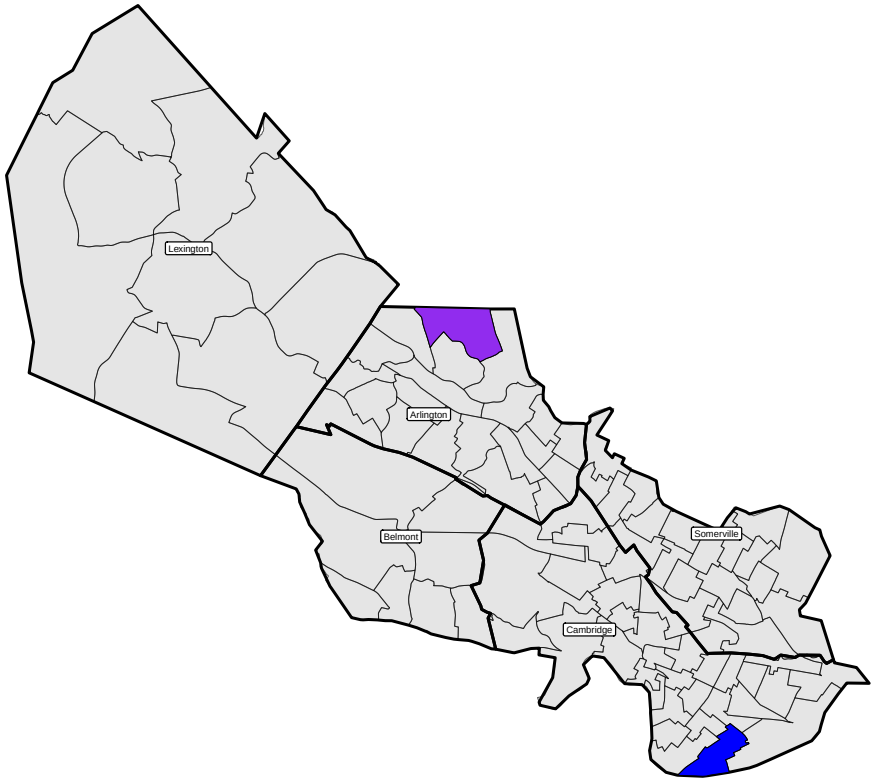

Figure 6: Highest (Blue) and Lowest (Purple) Democratic Vote Share

```
# ---- Moran's Testing Dem Vote Share ----
# ---- Step 1: Define neighboring voter precincts ----

#' construct a list of neighbors based on areas with contiguous boundaries
nb <- poly2nb(local_votes, queen = TRUE)

#' We can plot the neighbors of a given area
#' by adding a new column in map representing the neighbors of the area
cards <- card(nb) # number of neighbors per tract
max_id <- which.max(cards) # tract with the most neighbors

local_votes$neighbors <- "other"
local_votes$neighbors[max_id] <- "area"
local_votes$neighbors[nb[[max_id]]] <- "neighbors"
```

```r
ggplot(local_votes) +
  geom_sf(aes(fill = neighbors)) +
  scale_fill_manual(values = c("red2", "green2", "white")) +
  theme_void()
```

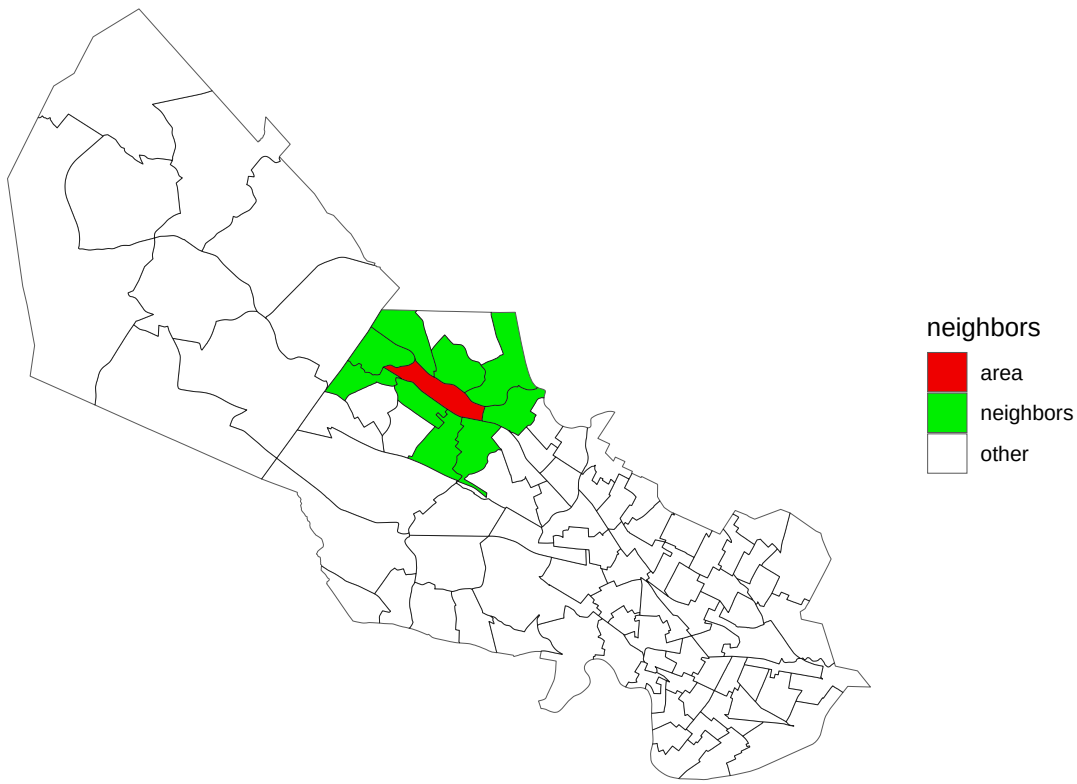

Figure 7: Voter Precinct with the Most Neighboring Precinct (9)

```r
# ---- Step 2: Assign weights to the neighbors ----

#' To compute a spatial weights matrix based on a binary neighbor list,
nbw <- nb2listw(nb, style = "W", zero.policy = TRUE)

# ---- Step 3: Compute the weighted neighbor mean income values ----

#' Compute the average neighbor DVS for each precinct
#' These are the spatially lagged values
```

```r
#' Spatial Lag
Vx <- lag.listw(nbw, local_votes$dvoteshare)
df <- data.frame(dvs = local_votes$dvoteshare, lag_dvs = Vx)

ggplot(df, aes(x = dvs, y = lag_dvs)) +
  geom_point(shape = 19) +
  geom_smooth(method = "lm", color = "red", se = TRUE) +
  labs(x = "Democratic Vote Share (DVS)",
       y = "Spatial Lag of DVS"
  ) +
  theme_minimal()
```

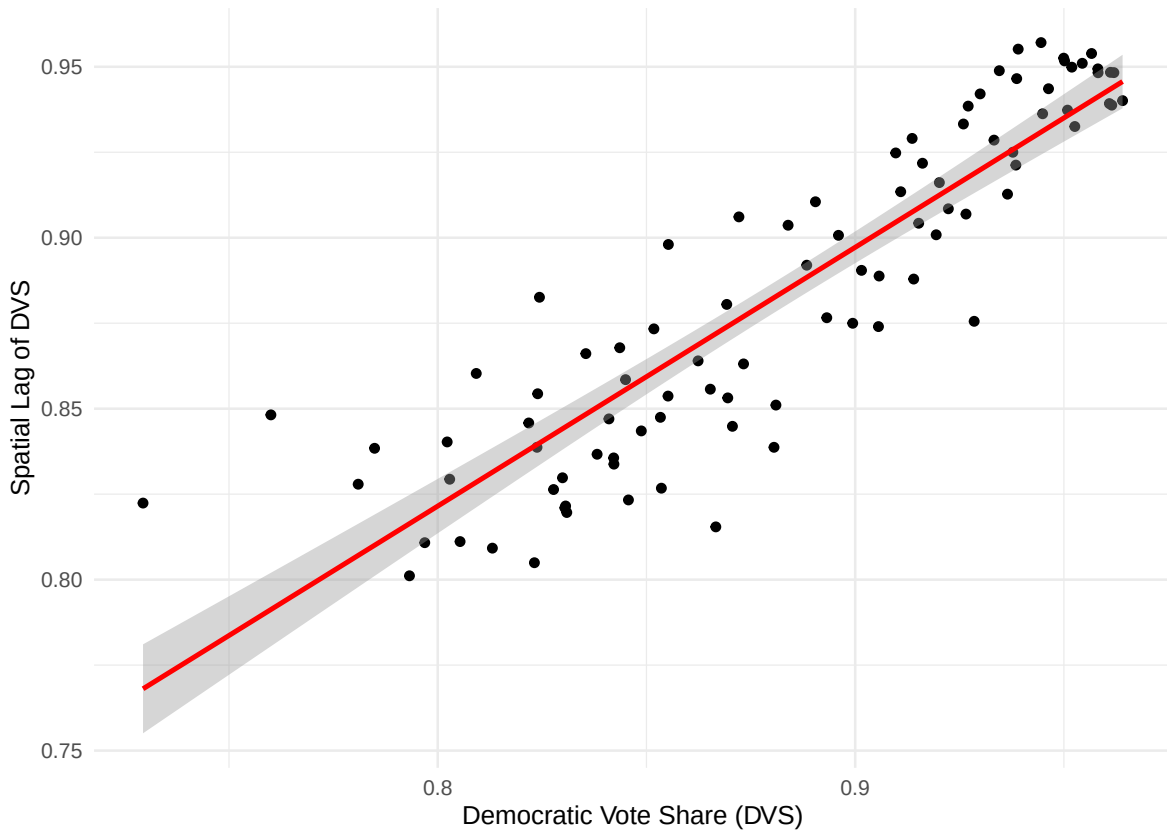

Figure 8: Association between the Dem Vote Share of Precinct and the DVS of Neighboring Precincts.

```r
# Fit the linear model
model2 <- lm(lag_dvs ~ dvs, data = df)

# ---- Step 4: Computing the Moran's I statistic ---
moran2 <- moran.test(local_votes$dvoteshare, nbw)
moran2
```

```
    Moran I test under randomisation

data:  local_votes$dvoteshare
weights: nbw

Moran I statistic standard deviate = 11.974, p-value < 2.2e-16
alternative hypothesis: greater
sample estimates:
Moran I statistic       Expectation          Variance
      0.756945540      -0.010989011       0.004113391
```

- Moran I estimate for Democratic vote share (DVS) than median household income (MHI), 0.7569 compared to 0.4123. The DVS of a precinct is far more positively associated with the DVS of its neighboring precincts than MHI is for precincts. This suggests that voting behavior (DVS) is more spatially clustered and locally homogeneous than economic status (MHI) and suggests that in the sample area, if people in neighboring precincts do not earn similarly, they still tend to vote similarly.

Question 4

Below is a map illustrating the locations of Whole Foods grocery stores in nearby towns. Note that it also includes two major roads in the area, Massachusetts Avenue and Alewife Brook Parkway (`mainroads`).

```
# Load Whole Foods in Middlesex County
# Requires: tidygeocoder, glue (used via :: below)
wholefoods <- tidygeocoder::geo(
  address =
    glue::glue("Whole Foods Market, {local_boundary %>% pull(NAME)}, Middlesex County, MA"
    full_results = TRUE,
    limit = 100) %>%
  filter(!is.na(lat)) %>%
  st_as_sf(coords = c("long", "lat"), crs = st_crs(local_tracts))

ggplot() +
  geom_sf(data = local_tracts, fill = NA, color = "lightgray") +
  geom_sf(data = local_boundary, color = "red", fill = NA, linewidth = 1) +
  geom_sf(data = mainroads, color = "orange") +
  geom_sf(data = wholefoods, color = "darkgreen", shape = 2) +
  geom_sf_label(data = local_boundary, aes(label = NAME),
                size = 1.5,
                color = "black",
                fill = "white") +
  theme_void()
```

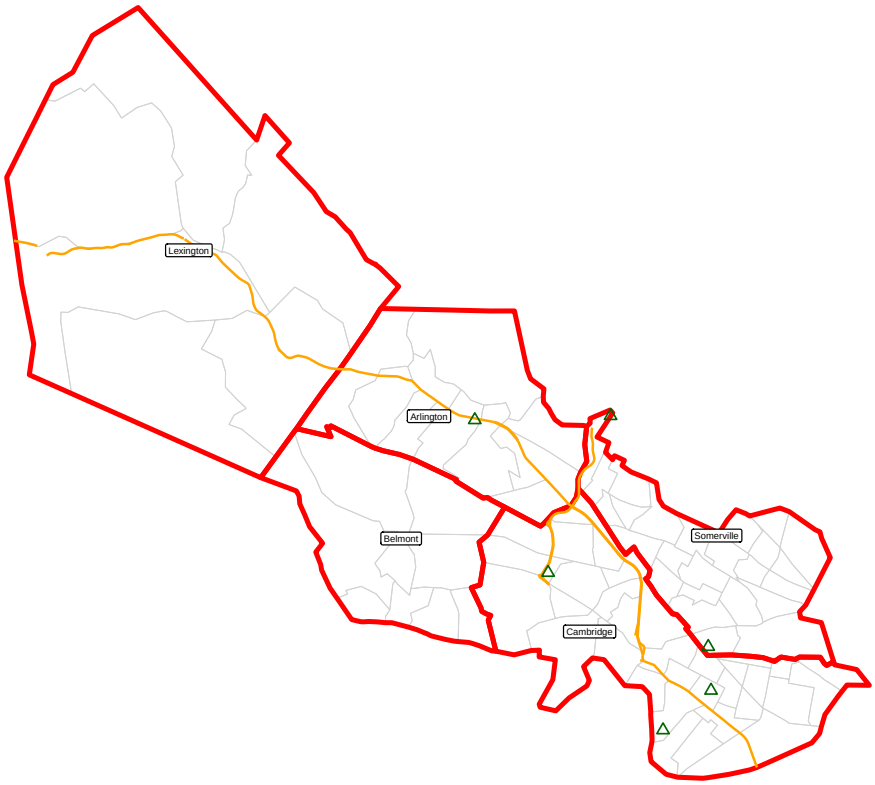

Figure 9: Local Whole Foods in Towns with Major Roads

Use `st_join()` to determine which local tracts have a Whole Foods. Compare the tracts with a Whole Foods to those without a Whole Foods on the following dimensions: (1) distance of the centroid of the tract from a major road; (2) population density; (3) household median income. What do these comparisons suggest about where Whole Foods chooses to locate?

Hint: Recall that you can use spatial functions `st_centroid()`, `st_area()`, and `st_distance()`.

Answer 4

```
# census tracts with a Whole Foods

# Note that in general the command
# data %>%
#   mutate(centroid = st_centroid(geometry))
# will create a column with the coordinates for the centroid
```

```r
# (a point towards the middle of a shape)
# for each row

Q4 <- st_join(local_tracts, wholefoods) %>%
  mutate(WF = if_else(!is.na(place_id), 1, 0))

Q4 <- Q4 %>%
  mutate(centroid = st_centroid(geometry),
         dist = st_distance(centroid, mainroads),
         pop_density = pop / (as.numeric(st_area(.)) / 1e6))

ggplot(Q4) +
  geom_sf() +
  geom_sf(data = local_boundary, color = "grey20", fill = NA, linewidth = 0.3) +
  geom_sf(data = Q4 %>% filter(WF == 1), fill = "grey70") +
  geom_sf(data = mainroads, color = "orange") +
  geom_sf(data = wholefoods,
        color = "green4", size = 2, shape = 18) +
  theme_urbn_map(
    scale = "discrete",
    base_size = 8.5,
    base_family = "sans",
    base_line_size = 0.5,
    base_rect_size = 0.5
  )
```

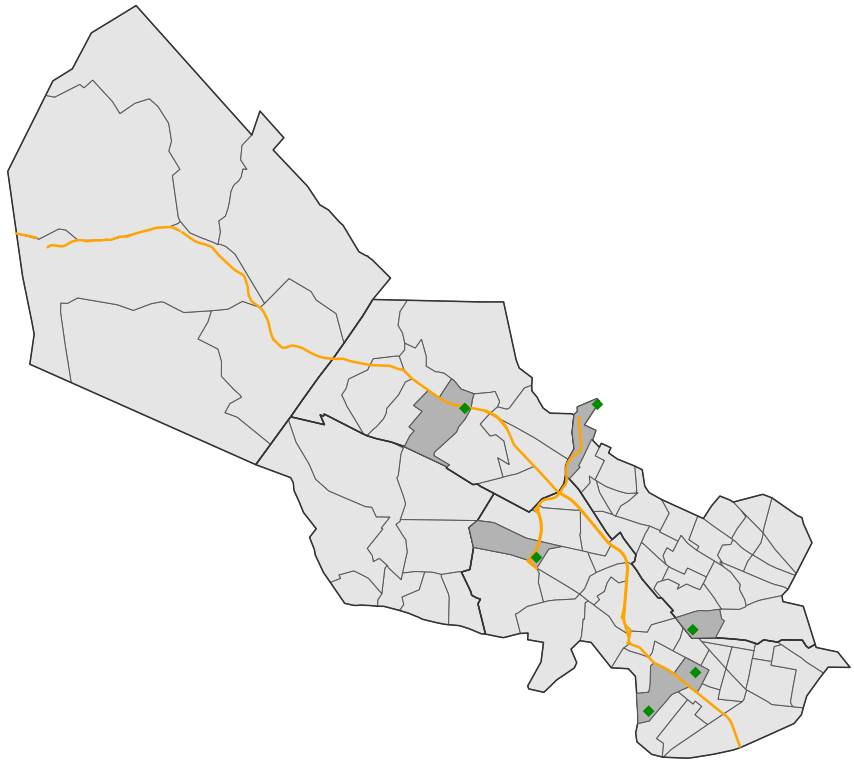

Figure 10: Cambridge-Area Census Tracts with Whole Foods

```
Q4a <- Q4 %>%
  filter(WF == 1)

Q4b <- Q4 %>%
  filter(WF == 0)
```

- For tracts with a Whole Foods in the Greater-Cambridge Area, the average distance of the tract centroid from a major road is 470, the population density is population density 7631, and the household median income is 1.13839×10^5.
- For all other tracts, the average distance of the tract centroid from a major road is 1180, the population density is population density 6662, and the household median income is 1.36834×10^5.

Statewide Analysis

Question 5

Now, we will expand our analysis to the whole state. Compare the vote share of precincts with Whole Foods to the vote share of precincts with Walmarts in Massachusetts. Also compare the median household income of census tracts with Whole Foods to those with Walmarts. What does this comparison reveal about the relationship between retail geography, income, and political preferences? See the code and map below as a start.

```r
# Download locations of all Walmarts and Whole Foods in MA
# Requires: tidygeocoder (used via :: below)
stores <- tidygeocoder::geo(
  address =
    c("Walmart, MA", "Whole Foods Market, MA"),
  full_results = TRUE,
  limit = 1000) %>%
  filter(!is.na(lat), class == "shop") %>%
  st_as_sf(coords = c("long", "lat"),
           crs = st_crs(local_tracts)) %>%
  st_filter(st_union(tracts)) %>%
  mutate(
    store = ifelse(str_detect(address, "Walmart"),
                   "Walmart", "Whole Foods"))

ggplot() +
  geom_sf(data = st_union(tracts), fill = NA) +
  geom_sf(data = stores, aes(color = store, shape = store)) +
  scale_color_manual(
    values = c("Whole Foods" = "darkgreen", "Walmart" = "dodgerblue")
    ) +
  labs(color = "Store", shape = "Store") +
  theme_void() +
  theme(legend.position = "bottom")
```

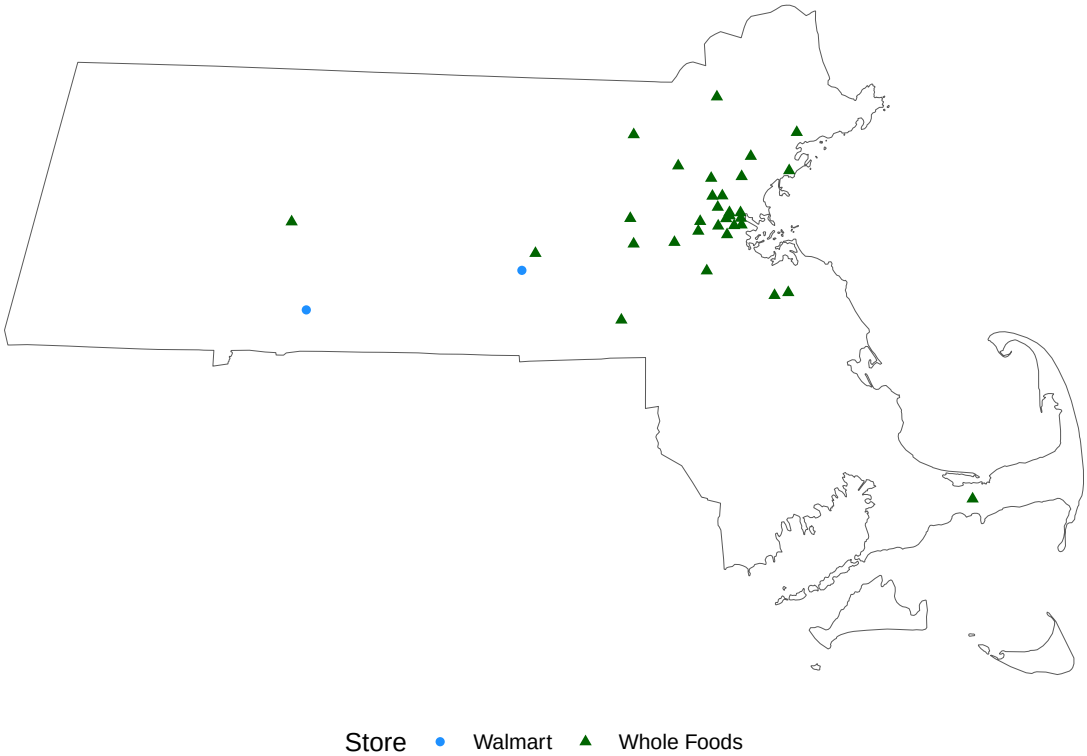

Figure 11: Massachusetts Whole Foods and Walmart Locations

Answer 5

```
sf3 <- tracts %>%
  st_join(stores)

# plot(st_geometry(sf3))

ggplot() +
  geom_sf(data = sf3, fill = NA, color = "grey80") +
  geom_sf(data = sf3 %>% filter(store == "Walmart"),
          color = "dodgerblue", fill = "dodgerblue") +
  geom_sf(data = sf3 %>% filter(store == "Whole Foods"),
          color = "darkgreen", fill = "darkgreen") +
  theme_void()
```

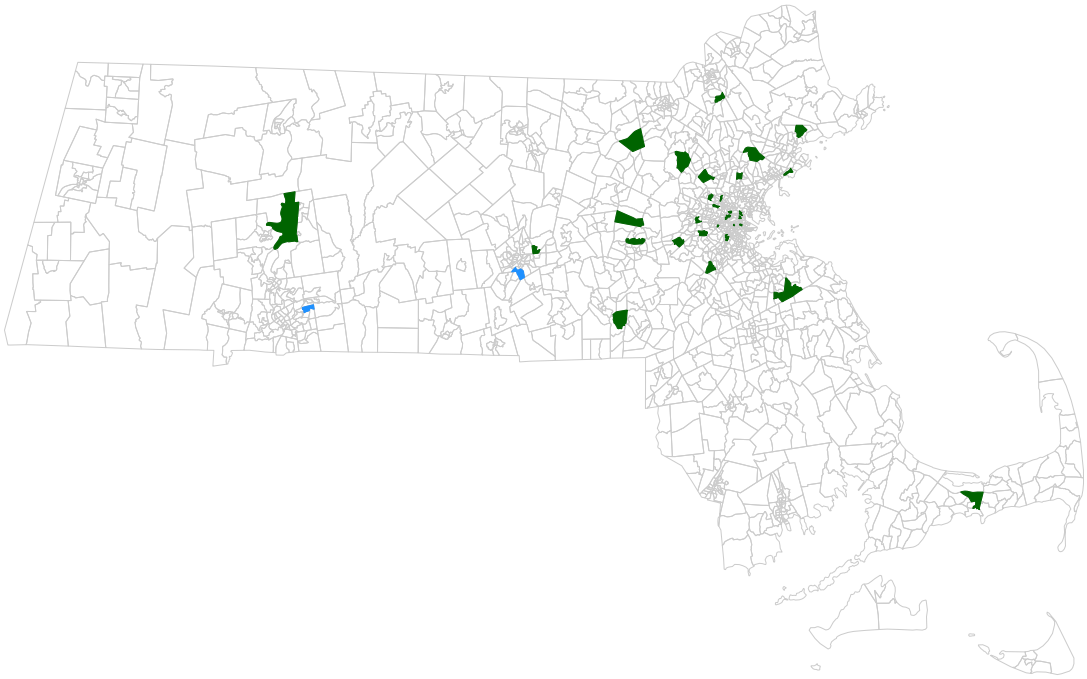

Figure 12: Massachusetts Census Tracts with a Whole Foods (Green) or Walmart (Blue)

```
#' Compare the vote share of precincts with Whole Foods to the DVS of
#' precincts with Walmarts in Massachusetts.

sf4 <- votes %>%
  st_join(stores)

ggplot() +
  geom_sf(data = sf4, fill = NA, color = "grey80") +
  geom_sf(data = sf4 %>% filter(store == "Walmart"),
          color = "dodgerblue", fill = "dodgerblue") +
  geom_sf(data = sf4 %>% filter(store == "Whole Foods"),
          color = "darkgreen", fill = "darkgreen") +
  theme_void()
```

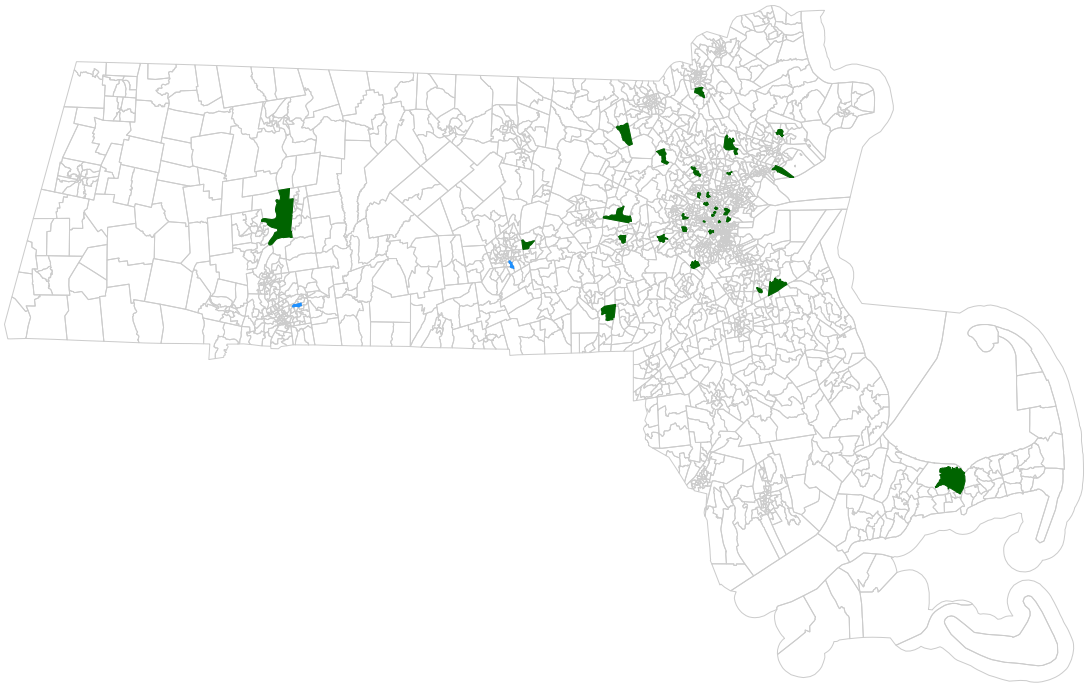

Figure 13: Voter Precincts with a Whole Foods (Green) or Walmart (Blue)

```
Q5c <- sf4 %>%
  filter(store %in% c("Walmart", "Whole Foods")) %>%
  group_by(store) %>%
  summarize(avg_dvs = mean(dvoteshare)) %>%
  st_drop_geometry()

Q5c %>% kbl()
```

store	avg_dvs
Walmart	0.7279544
Whole Foods	0.7714898

Average DVS of Voter Precincts with a Whole Foods (Green) or Walmart (Blue)

```r
t.test(dvoteshare ~ store,
       data = sf4 %>%
                filter(store %in% c("Walmart", "Whole Foods")) %>%
                st_drop_geometry())
```

```
    Welch Two Sample t-test

data:  dvoteshare by store
t = -1.6722, df = 7.5542, p-value = 0.1353
alternative hypothesis: true difference in means between group Walmart and group Whole Foods
95 percent confidence interval:
 -0.10419398  0.01712327
sample estimates:
    mean in group Walmart mean in group Whole Foods
                0.7279544                 0.7714898
```

```r
#' Compare the MHI of census tracts with Whole Foods to those with Walmarts

Q5d <- sf3 %>%
  filter(store %in% c("Walmart", "Whole Foods")) %>%
  group_by(store) %>%
  summarize(avg_mhi = mean(inc)) %>%
  st_drop_geometry()

Q5d %>% kbl()
```

store	avg_mhi
Walmart	61841.5
Whole Foods	116341.8

Average MHI of Voter Precincts with a Whole Foods (Green) or Walmart (Blue)

```r
t.test(inc ~ store,
       data = sf3 %>%
                filter(store %in% c("Walmart", "Whole Foods")) %>%
                st_drop_geometry())
```

```
    Welch Two Sample t-test
```

```
data:  inc by store
t = -6.5373, df = 17.53, p-value = 4.357e-06
alternative hypothesis: true difference in means between group Walmart and group Whole Foods
95 percent confidence interval:
 -72049.11 -36951.52
sample estimates:
    mean in group Walmart mean in group Whole Foods
                  61841.5                  116341.8
```

- The Democratic vote share (DVS) of Massachusetts precincts with Whole foods is 0.771 compared to the DVS in Walmart precincts of 0.728, meaning that Whole Foods precincts have a DVS approximately 4.4% higher than Walmart precincts. There is some evidence that Whole Foods precincts lean more Democratic than Walmart precincts but the difference is not statistically significant (t-test $p = 0.1353$). This indicates that, at least in Mass, there is a negligible political difference (expressed as Dem vote share) between voter precincts with Walmarts and those with Whole Foods.
- The median household income (MHI) of Mass census tracts with Whole Foods is $116,342 compared to $61,841.50 in Walmart tracts, meaning that Whole Foods precincts have a MHI approximately $$54,500.31. This difference is substantively and statistically significant (t-test $p < 0$) indicating that there is a correlation between higher median household income and the location of a Whole Foods (relative to Walmart).

Redistricting

Question 6 (Optional Challenge Question)

What is the maximum number of congressional districts you can draw in Massachusetts that are Republican based on 2020 two-party Presidential election vote share? Use the shortburst algorithm with a burst size of 20 and 1000 bursts to find an answer.

What is the Democratic vote share in the *most Republican* district that you can draw? What is the compactness in terms of Polsby-Popper of the least compact district?

Answer 6

```
# Insert Answer Here

# Hint to get started
# First we create a Redist Map Object
ma <- redist_map(
  votes %>%
    mutate(
      tot = pre_20_dem_bid + pre_20_rep_tru
      ),
  pop_tol = .01,
  total_pop = "pop",
  ndists = 9,
  planarize = st_crs(votes))

# Use the following seed for reproducibility
set.seed(4)

# Now the next step is to run shortburst as in class lab solutions!
```